

Reactive Skin and Joint Conditions in CSD

Reactive conditions involving the skin and joints can occur in patients with cat scratch disease (CSD).

- **Erythema nodosum** develops in approximately **2%** of CSD patients.
- **Rheumatoid factor-negative arthropathy** affects about **3%** of patients, typically involving the large joints of the extremities.
- These patients are usually **over 20 years of age**, with **two-thirds being female**.
- **20%** of these arthritic patients also have erythema nodosum.
- In **80%** of cases, the arthropathy resolves within **6 weeks**; however, **20%** experience a more **chronic course**, averaging **2.5 years** in duration.

Laboratory Tests

Routine laboratory findings in CSD are **nonspecific** and primarily useful for excluding other diagnoses.

- During the acute phase, patients may exhibit:
- Mildly **elevated white blood cell count**
- **Elevated erythrocyte sedimentation rate (ESR)**
- **Blood cultures** are rarely positive with current techniques.
- **PCR testing** of affected lymph nodes is positive in approximately **30%** of cases.
- **Serology for *Bartonella henselae*:**
- Positive in **80–90%** of cases at presentation.
- Many initially seronegative patients become positive within **2 to 8 weeks**.
- **Serologic testing** has replaced the CSD skin test as the **preferred confirmatory method**.

Histopathology

Histopathologic examination of lymph nodes in CSD shows nonspecific changes with three overlapping stages of granuloma formation:

1. **Lymph node enlargement** with germinal center hypertrophy and cortical thickening.
2. **Granuloma formation** with infiltration by lymphocytes and epithelioid cells.
3. **Necrosis** accompanied by neutrophil infiltration.

These stages may coexist within a single lymph node.

- **Abscess formation** may also occur.

- *B. henselae* organisms can be identified in lymph nodes and primary skin lesions, often in **clumps or filamentous forms**, typically located **within or near areas of necrosis**.

Primary Skin and Conjunctival Lesions

- **Skin lesions** show:
 - Necrosis of the epidermis and upper dermis
 - Dermal inflammation ranging from neutrophil and macrophage infiltration to granuloma formation
- **Conjunctival lesions** exhibit similar features.
- In some cases, there is vascular proliferation and a **homogeneous eosinophilic to basophilic granular background**, resembling **bacillary angiomatosis (BA)**.

Differential Diagnosis

Most Likely

- Atypical Mycobacteria (especially *M. marinum*)
- Sporotrichosis

Consider

- Syphilis
- Tularemia

- Primary inoculation tuberculosis

Rule Out

- Lymphoma

Treatment

- **Most cases of CSD resolve spontaneously** without treatment.
- A retrospective study of 202 patients showed that commonly used antibiotics (macrolides, tetracyclines, cephalosporins) were **ineffective** in classic CSD.
 - Despite *in vitro* susceptibility of *B. henselae* to macrolides, tetracyclines, and second- and third-generation cephalosporins, clinical benefit in immunocompetent hosts is limited—likely due to low infectious burden and late initiation of therapy, often after spontaneous improvement has begun.

Azithromycin

- In a prospective randomized trial, **5-day azithromycin** treatment:
 - Reduced lymph node volume during the first **30 days**
 - **50%** of treated patients reached **20% of baseline volume** by 1 month vs. **7%** in untreated
 - No significant difference in other symptoms or long-term outcomes
 - Some patients experienced **lymph node enlargement after treatment**

Complicated CSD

For **ocular** or **central nervous system** involvement:

- Recommended regimen:
 - **Doxycycline 100 mg twice daily**
 - **Rifampin 300 mg twice daily**

Trench Fever, Urban Trench Fever, and *Bartonella quintana* Infections

Epidemiology

- *Bartonella quintana* is the causative agent of **trench fever**, which affected over **1 million soldiers** during **World War I**.
- In recent decades, *B. quintana* has emerged as a cause of **bacteremia** and **endocarditis**, particularly in urban populations—termed **urban trench fever**—reported in the **United States and France**.